

Sumit Singh Bagdwal

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EDUCATION

Graphic Era Hill University

B.Tech in Computer Science and Engineering | CGPA: 8.26/10

Bhimtal, Uttarakhand

2024 – 2028

TECHNICAL SKILLS

Languages: C++, JavaScript, SQL

Frontend: React.js, Tailwind CSS, HTML, CSS

Backend: Node.js, Express.js, REST APIs, WebSockets, Socket.IO

Databases: MySQL, MongoDB

Tools & Platforms: Git, GitHub, Docker, Postman, Vercel, Railway, VS Code

Core CS: Data Structures & Algorithms, OOP, DBMS, System Design

PROJECTS

IBVAP – Intelligent Border Video Analytics Platform

[GitHub](#)

React.js, Node.js, FastAPI, MySQL, Redis, YOLO11, ByteTrack, OpenCV, Docker, AWS EC2

- Built and deployed a real-time video analytics platform using **YOLO11**, **ByteTrack**, **ANPR/EasyOCR**, and **YuNet** to detect and track people, vehicles, license plates, and faces, with inference accelerated using **Apple MPS**.
- Developed a context-aware risk system with dynamically configurable surveillance zones, virtual fences, and risk rules to evaluate object movement, support flexible threat policies, and maintain a single evolving incident per tracked object.
- Deployed the application on **AWS EC2** with **Docker**, **Nginx**, **Node.js**, **MySQL**, and **Redis**, while running the **FastAPI AI engine** on an edge device connected securely through **Tailscale**, with automatic RTSP reconnection and offline handling.

JanaSahaya – Civic Issue Reporting & Resolution Platform

[GitHub](#)

React.js, Node.js, Express.js, MySQL, Socket.IO, Docker, Vercel, Railway

- Built a **full-stack civic issue reporting platform** for **3 user roles**—citizens, officers, and administrators—using a **22-table MySQL database**, with features for geotagged complaints, issue assignment, resolution tracking, SLA monitoring, analytics, and audit logs.
- Developed a **0–100 priority scoring system** and duplicate-issue detection using **Haversine distance**, **text/category similarity**, **issue age**, **votes**, **nearby complaints**, and **escalation signals** to help prioritize reported issues.
- Implemented **JWT authentication**, **RBAC**, **MySQL transactions**, and **Socket.IO real-time updates**, and verified the platform with **43 automated tests**; documented **54 API operations**, and deployed the frontend on Vercel with the Dockerized backend and MySQL on Railway.

ACHIEVEMENTS & CERTIFICATIONS

- **Shortlisted for Smart India Hackathon 2026** with the IBVAP intelligent border surveillance project.
- Finalist in **Hack the Winter 2025**, a national-level hackathon.
- Achieved the **Google Cloud Arcade Legend Tier** through hands-on cloud labs and technical skill challenges.

LEADERSHIP & ACTIVITIES

Lead, AWS Student Builder Group

GEHU Bhimtal

Student Technical Community

2026 – Present

- Lead student learning initiatives in **software development**, **cloud computing**, **data engineering**, and **problem solving**.
- Organize **coding sessions**, **technical workshops**, **learning challenges**, and **campus technology events** for the student developer community.

National Cadet Corps (NCC) | NCC B Certificate